

19-205-0802 Materials Management

Module 3

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- **Dynamic inventory problems under risk** deal with managing inventory over multiple time periods.
- key elements are demand, supply lead times, prices, or costs
- Which are uncertain **and decision-makers are risk-averse**, not just profit-maximizing.

- Decisions are made sequentially over time (period by period).
- Current inventory decisions affect future states and costs.
- Typically modeled using dynamic programming (DP) or stochastic control.
- Dynamic Programming is a mathematical optimization framework used to solve **multi-stage decision problems**, where decisions are made sequentially over time under uncertainty.

Stochastic Control

- Stochastic control generalizes DP by explicitly modeling uncertainty and risk preferences in continuous or discrete time.

Sources of risk in inventory systems

- Demand uncertainty (most common)
- Supply disruptions or variable lead times
- Price volatility (purchase or selling price)
- Holding and shortage cost uncertainty

Risk-neutral vs risk-averse inventory models

- **Risk-neutral models**
- Objective: minimize *expected total cost* or maximize *expected profit*
- Example: classic stochastic inventory control, policies
- **Risk-averse models**
- Objective: control **variability or downside risk**, not just expectation
- Decision-maker prefers *stable outcomes* over high but risky profits

Common risk measures used

- Dynamic inventory problems under risk often incorporate:
- **Variance of cost/profit**
- **Mean–variance trade-off**
- **Value-at-Risk (VaR)**
- **Conditional Value-at-Risk (CVaR)**
- **Utility functions** (exponential, concave utilities)

Optimal policies

- Risk aversion often leads to:
 - **Higher safety stock**
 - **More conservative ordering**
 - **Lower probability of stockouts**

Solution approaches

- Stochastic dynamic programming
- Approximate DP (for large state spaces)
- Scenario-based optimization
- Robust optimization
- Simulation-based methods

Practical applications

- Spare parts management
- Pharmaceutical and medical inventory
- Retail and e-commerce supply chains
- Humanitarian logistics
- Energy and fuel inventory systems

General Characteristics

1. Sequential decision-making

- Decisions are made **period by period** based on the current system state.
- Each decision affects **future inventory levels and costs**.

2. State-dependent policies

- Optimal decisions depend on the **current state** (e.g., inventory level, demand history).
- Policies are typically expressed as **state–action mappings**.

3. Uncertainty modeling

- Randomness in demand, supply, or lead time is explicitly modeled.
- System evolution follows a **stochastic transition law**.

4. Risk incorporation

- **DP** often assumes risk neutrality (expected cost minimization).
- **Stochastic control** allows **risk-averse objectives** using utility functions, variance penalties, VaR, or CVaR.

5. Bellman optimality principle

- Optimal decisions satisfy the principle that **future actions remain optimal regardless of past decisions.**
- Leads to recursive solution equations (Bellman or HJB equations). Hamilton–Jacobi–Bellman -optimality equations

6. Policy structure

- Many inventory systems yield **simple structural policies**:
 - (s, S) , base-stock, or threshold policies
- Risk aversion generally increases **safety stock levels.**

7. Time horizon flexibility

- Applicable to **finite or infinite horizons**.
- Can be formulated in **discrete or continuous time**.

8. Computational complexity

- Suffers from the **curse of dimensionality** as state space grows.
- Approximate and simulation-based methods are often required.

9. Performance trade-offs

- Balances **expected cost** against **risk or variability**.
- Conservative ordering policies reduce stockouts but may increase holding costs.

10. Practical relevance

- Widely used in supply chains, manufacturing, spare parts systems, healthcare, and energy inventory management.

Basic Kinds of Inventory Control Systems

1. Continuous Review System (Q-system)

- Inventory level is **monitored continuously**.
- An order of **fixed quantity Q** is placed when inventory falls to a **reorder point RRR**.
- Also known as the **(Q,R)** or **(s,Q)** system.

Characteristics:

- Suitable for **high-value or critical items**
- Lower safety stock requirement
- Requires continuous monitoring
- **Example:** Spare parts, industrial components

2. Periodic Review System (P-system)

- Inventory is reviewed at **fixed time intervals T**.
- Order quantity varies to raise inventory up to a **target level S**.
- Known as **(T,S)** or **(P,S)** system.
- **Characteristics:**
 - Easier to administer
 - Higher safety stock required
 - Suitable for **low-cost items**
- **Example:** Retail goods, grocery items

3. Base Stock System

- A special case of continuous review.
- Each demand triggers an order so that inventory is restored to a **base-stock level SSS**.
- **Characteristics:**
- Common in **make-to-stock** environments
- Simple structure
- Effective under stationary demand

4. Order-up-to Level System

- Used mainly in periodic review.
- Order quantity = difference between **order-up-to level** **SSS** and current inventory.
- **Characteristics:**
- Flexible ordering
- Widely used in supply chains

5. Two-Bin System

- Inventory is stored in **two separate bins**.
- When the first bin is empty, an order is placed.
- **Characteristics:**
- Very simple and visual
- No detailed record keeping
- Suitable for **low-value items**

6. Just-In-Time (JIT) System

- Inventory is replenished **only when needed**.
- Aims to minimize inventory levels.
- **Characteristics:**
- Requires reliable suppliers
- Low tolerance for demand variability
- Emphasizes waste reduction

7. Material Requirements Planning (MRP)

- Inventory control based on **production schedules and bills of materials.**
- **Characteristics:**
- Deterministic demand
- Suitable for manufacturing systems

General characteristics

- **1. Multi-period decision making**

Inventory decisions (order quantity, timing) are made repeatedly over multiple time periods, with each decision affecting future inventory levels and costs.

- **2. Uncertainty (risk) in system parameters**

Key elements such as customer demand, lead time, supply availability, or prices are random variables and not known with certainty at the time of decision making.

- **3. State-dependent evolution**

The system state (inventory level, backorders, pipeline stock) evolves dynamically over time according to previous decisions and realized random outcomes.

- **4. Sequential decision process**

Decisions are taken at discrete or continuous time intervals based on the current state of the system and available information.

- **5. Trade-off between costs and service level**

The objective typically balances ordering costs, holding costs, shortage or backorder costs, and sometimes penalty or service-level constraints under risk.

- **6. Risk exposure and variability**

Variability in demand or supply introduces the risk of stockouts or excess inventory, making variability control a central concern.

- **7. Stochastic modeling framework**

The problem is usually formulated using stochastic processes such as Markov decision processes (MDP), stochastic dynamic programming, or stochastic control models.

- **8. Optimal policy structure**

Solutions often yield structured policies (e.g., (s,S) , base-stock, or order-up-to policies) that specify optimal actions under uncertainty.

- **9. Expectation-based or risk-sensitive objectives**

Performance criteria may involve expected total cost, discounted cost, or risk-adjusted measures (e.g., variance, CVaR). Conditional value at risk.

- **10. Information updating over time**

As time progresses, new demand and supply information becomes available, allowing decisions to adapt dynamically.